Tutorial- 04

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1) **What is wrong with the following if statement (there are at least 3 errors). The Indentation indicates the desired behavior.**

```
if numNeighbors >= 3 || numNeighbors = 4
++numNeighbors;
 printf("You are dead! \n " );
  else
--numNeighbors;
```

- ```
  if ( numNeigbors >= 3 || numNeighbors == 4 );
  ++numNeighbors ;
  printf( " You are dead! \n ") ;
  else
     --numNeighbors ;
  ```

2) **Describe the output produced by this poorly indented program segment:**

```
int number = 4;
double alpha = -1.0;
if (number > 0)
if (alpha > 0)
printf("Here I am! \n" );
else
```
```

```
printf("No, I'm here! \n");
printf("No, actually, I'm here! \n");
```

- In this program the data type of 4 value is integer and its variable name is number . In the next line it shows value -1.0 it's variable name is alpha and data type is double. In if statement line number 4 is greater than the 0 so it is true but next if alpha is -1.0 greater than 0 is false . because of that it goes to the else statement and print output as ,

  No, I'm here!

  No, actually, I'm here !

3)Consider the following if statement, where doesSignificantWork, makesBreakthrough, and nobelPrizeCandidate are all boolean variables:

```
if (doesSignificantWork)
{
if (makesBreakthrough)
nobelPrizeCandidate = true;
else
 nobelPrizeCandidate = false;
 }
else if (!doesSignificantWork)
 nobelPrizeCandidate = false;
```

- If (doesSignificantWork) true, then go to the if (makesBreakthrough) if it is true then noblePrizeCandidate is true. If (makesBreakthrough) is false, then noblePrizeCandidate is False .
- If (doesSignificantWork) is false then noblePrizeCandidate is false.

4) Write if statements to do the following:

– If character variable tax Code is ’T’, increase price by adding the tax Rate percentage of price to it.

– If integer variable opCode has the value 1, read in double values for X and Y and calculate and print their sum.

– If integer variable currentNumber is odd, change its value so that it is now 3 times currentNumber plus 1, otherwise change its value so that it is now half of currentNumber (rounded down when currentNumber is odd).

– Assign true to the boolean variable leapYear if the integer variable year is a leap year. (A leap year is a multiple of 4, and if it is a multiple of 100, it must also be a multiple of 400.)

– Assign a value to double variable cost depending on the value of integer variable distance as follows:

Distance	Cost
----------------------------------	----------
0 through 100	5.00
More than 100 but not more than 500	8.00
More than 500 but less than 1,000	10.00
1,000 or more	12.00

- if taxCode == T ;
  price += price * taxRate / 100 ;
-